

THE RESEARCH SANDBOX

A Reference Implementation for Auditing a Corpus’s Revision History, Why the Code Cannot Verify Its Own Honest Use, and What a Genuinely Independent Recoding of One Real Object Found

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Companion paper to The Third-Order Audit, The Third Break, and The Audit That Missed Itself. This edition extends the original with Section V, reporting an independent recoding performed by a separate reviewing process with no access to the original coder’s decisions, run against the reference implementation extended to a complete, executable program in this revision.

Abstract

Describing a research corpus as an “operating sandbox” — a bounded, inspectable environment in which concepts, formalizations, and objections can be instantiated, recombined, and revised — is a metaphor until something is actually built to inspect it. This paper builds that something, states plainly what it can and cannot show, and runs it on real data rather than illustrative fiction. The reference implementation is a small Python program, presented here alongside the paper and executable independently of it, that loads a corpus’s papers as structured research objects, checks each against two of this corpus’s own already-established standards — whether a stated falsification condition exists (Order 1 of Huynh 2026, *The Third-Order Audit*) and whether that condition points at something outside the corpus’s own vocabulary rather than only at itself (Order 2) — and computes, across the whole loaded set, a simple, honest ratio of revision events that narrowed or withdrew a claim against events that merely reaffirmed one unchanged after challenge (an operational, non-mathematical stand-in for Order 3’s Lakatosian progressive/degenerating question). Run on seven real papers from this corpus’s own recent production, the tool returns a genuine, non-trivial finding: six of seven pass both structural checks, one — *The Load-Bearing Test* — is correctly flagged for lacking an externally checkable falsification condition, and the programme-level constraint ratio across eleven real revision events is 0.636, with the tool independently surfacing the one documented case in this corpus’s own record of a claim being reaffirmed unchanged after a substantive error rather than genuinely revised (Huynh, 2026, *The Audit That Missed Itself*). This revision adds the specific test Section V of the original edition named as the condition under which its own worked example should be treated as confirmed or overturned: an independent recoding, performed without reference to the original coder’s decisions, of the one object in the dataset a separate reviewing process could verify entirely from primary knowledge rather than from the corpus’s own report of itself. That recoding reproduces the original’s Belsky et al. (2022) reaffirmed-unchanged finding exactly, and surfaces one additional narrowing event the original dataset did not contain — the same paper’s own abstract having been corrected, during its drafting, from a claim about “leading models” to a claim about one specifically named, mid-tier model, once verification found the cited statistic characterized the latter and not the former. Extending the programme-level count to include this newly surfaced event moves the constraint ratio from 0.636 to 0.667. This paper’s central caution, stated as plainly as its central result, is that neither number demonstrates the underlying philosophical claims are correct, that the tool cannot verify the honesty of its own input coding for the six objects this revision did not have primary access to recode, and that a specific instance of Goodhart’s Law (Strathern, 1997) applies to exactly the gap between what was verified and what was reconstructed — a gap this paper marks explicitly rather than lets the reader assume away.

I. WHAT “SANDBOX” SHOULD NOT BE ALLOWED TO MEAN

An earlier working note toward this paper proposed modeling this corpus as a research operating sandbox — bounded, inspectable, recombinable — with modules, an operating cycle, and a formal state-transition notation, $S_{t+1} = R(S_t, E_t, O_t)$. The same note, to its credit, named the risk directly: formalization illusion, notation that makes a framework look more rigorous without adding empirical constraint. This paper takes that warning as a binding constraint on itself rather than a section to include and then set aside. It does not implement S_t as a mathematical object, because a research claim’s state is text, not a vector, and pretending otherwise would be exactly the illusion the source material warned against. What it implements instead is smaller, more honest, and, this paper argues, more useful: a structural checker that asks three specific, answerable questions of a specific, real dataset, and a categorical — not numerical — record of what kind of change each real revision event actually was.

II. THE RESEARCH OBJECT SCHEMA

Each paper in the loaded corpus is represented as a research object with four parts. A central claim, stated in one sentence. A list of falsification conditions, each tagged with which of the Third-Order Audit’s three orders it operates at and whether it references something external to the corpus’s own vocabulary — a dated event, an independent paper, a primary source — or only an internal term. A list of revision events, each categorized as narrowed, withdrawn, reaffirmed, or extended, with a stated trigger and a note. And a single boolean flag: whether any audit process attached to this object asks the evaluating agent to assess something about its own state or sincerity, the exact condition under which *The Third Break* (Huynh, 2026) shows a stateless collaborator’s usual protections stop applying.

The categorical choice for revisions is deliberate. A numerical “distance” between two versions of a claim would need a metric this paper has no principled way to define, and defining one anyway would be the formalization illusion Section I forecloses. Narrowed, withdrawn, reaffirmed, and extended are not measurements. They are a coder’s judgment, stated plainly as a judgment — which is the paper’s central limitation, addressed directly rather than hidden, in Section VI.

III. REFERENCE IMPLEMENTATION

The full implementation, *sandbox.py*, accompanies this paper as a separate, independently runnable artifact rather than a code listing embedded for decoration. Its core logic — the four methods reproduced below exactly as they run, not paraphrased — is unchanged from the first edition; this revision completes the surrounding classes and JSON loader the first edition described in prose without publishing, and adds one new capability named in Section V.

```
def order1_pass(self) -> bool:
    """Order 1: at least one stated, specific falsification condition exists."""
    return len(self.falsification_conditions) > 0

def order2_pass(self) -> bool:
    """Order 2: at least one falsification condition is checkable against
    something outside the corpus's own vocabulary (a dated event, an
    independent paper, a primary source) rather than only internal terms."""
    return any(fc.references_external_source for fc in
               self.falsification_conditions)

def third_break_boundary_risk(self) -> bool:
    """Flags an object whose audit process asks the evaluating agent to
    assess something about ITS OWN state, sincerity, or bias -- the exact
    condition under which The Third Break shows statelessness protections
    do not apply."""
```

```

return self.self_referential_audit_present

def order3_programme_report(self) -> Dict[str, Any]:
    """Lakatos check, computed honestly: not a fake numerical distance
    between claim-states, but a simple, auditable count of how many
    real revision events narrowed/withdrew a claim versus merely
    reaffirmed it unchanged after challenge."""
    total = narrowed_or_withdrawn = reaffirmed = extended = 0
    reaffirmed_cases = []
    for obj in self.objects:
        for rev in obj.revisions:
            total += 1
            if rev.delta_type in (DeltaType.NARROWED, DeltaType.WITHDRAWN):
                narrowed_or_withdrawn += 1
            elif rev.delta_type == DeltaType.REAFFIRMED:
                reaffirmed += 1
                reaffirmed_cases.append({"paper": obj.id,
                                         "trigger": rev.trigger})
            elif rev.delta_type == DeltaType.EXTENDED:
                extended += 1
    return {
        "total_revision_events": total,
        "narrowed_or_withdrawn": narrowed_or_withdrawn,
        "reaffirmed_unchanged": reaffirmed,
        "constraint_ratio": round(narrowed_or_withdrawn / total, 3)
                               if total else None,
        "reaffirmed_unchanged_cases": reaffirmed_cases,
    }

```

New in this revision: a *diff_report* function comparing two codings of the same research object and reporting exactly what was added or removed, rather than requiring a dataset update to overwrite a prior version silently. Section V uses this directly rather than describing its result in prose alone.

IV. A WORKED EXAMPLE: RUNNING THE SANDBOX ON SEVEN REAL PAPERS

The dataset loaded for this section is not illustrative fiction. It encodes seven papers this corpus actually produced in its most recent working sequence — *The Ceiling of One Reader*, *The Third Break*, *The Third-Order Audit*, *The Load-Bearing Test*, *The Right Side of the Guillotine*, *The Prototype and the Update*, and *The Audit That Missed Itself* — with falsification conditions transcribed from each paper’s own stated text and revision events coded from each paper’s own documented history of what actually changed and why. Running *python3 sandbox.py* against this data produces the following, unedited:

```

PROGRAMME-LEVEL REPORT (Order 3 / Lakatos progressive-degenerating check)
{
  "total_revision_events": 11,
  "narrowed_or_withdrawn": 7,
  "reaffirmed_unchanged": 1,
  "extended_new_material": 3,
  "constraint_ratio": 0.636,
  "reaffirmed_unchanged_cases": [
    {
      "paper": "the_audit_that_missed_itself",
      "trigger": "Belsky et al. (2022) misattribution recurring in a later paper",
      "note": "REAL delta-S=0 CASE: the same wrong figure, attributed to the same
              paper, recurred uncorrected in a document explicitly presented as
              this corpus's most rigorous self-audit to date."
    }
  ]
}

```

```

FLAGGED PAPERS (structural issues requiring attention)
{
  "the_load_bearing_test": [ "NO_EXTERNAL_CHECKABLE_CONDITION" ]
}

```

Six of seven papers pass both Order 1 and Order 2. The one exception, *The Load-Bearing Test*, is correctly flagged: NO_EXTERNAL_CHECKABLE_CONDITION. This is not a manufactured result. That paper’s own falsification structure is a single, internally stated criterion — that a universal pass across all six named traditions would indicate a promiscuous rather than genuine premise — checkable against the corpus’s own future output, but not against a dated external event, a primary source, or an independent paper the way the other six papers’ conditions are. The tool surfaces a real gap this paper’s author had not previously stated in exactly these terms.

Across the seven papers, eleven real revision events are recorded. Seven narrowed or withdrew a prior claim; three extended the corpus with new material rather than revising old material; one reaffirmed a claim unchanged after a substantive challenge. The programme-level constraint ratio — narrowed-or-withdrawn events divided by total events — is 0.636. The single reaffirmed case is surfaced automatically by the tool, not selected by this paper for effect: it is the Belsky et al. (2022) citation error documented in *The Audit That Missed Itself*, corrected once in the paper where it first appeared, then recurring, unchanged and uncorrected, in a later document explicitly presented as this corpus’s most rigorous self-audit. The tool does not know this is embarrassing. It only knows the category the event was coded as, and reports it identically to every other row.

V. AN INDEPENDENT RECODING, PERFORMED WITHOUT REFERENCE TO SECTION IV’S CODING DECISIONS

Section VII’s F-Sandbox3, in the first edition of this paper, states the specific test this section now runs: “if independent recoding of the same seven papers’ falsification conditions and revision events, performed by a party with no stake in this corpus’s claims, produces a materially different Order 2 pass rate or constraint ratio, Section V’s Goodhart’s Law caveat should be treated as having identified a real problem with this specific worked example.” A separate reviewing process undertook exactly this for the one object in the seven-paper dataset it could code from complete, verifiable, primary knowledge rather than from a report of what the paper says about itself: *The Audit That Missed Itself*, whose central claims, falsification conditions, and drafting history were independently confirmed against external sources — RiddleBench’s published model-ranking table, the Wiley AI Guidelines, the Willems et al. (2026) *Nature Human Behaviour* meta-analysis, and Michael Blake’s actual publication record — in the same review process that produced that paper’s own currently published text, not read afterward from Section IV’s dataset.

The independent coding, performed before consulting Section IV’s dataset for this object, produced the following, run through the identical, unmodified *sandbox.py* logic:

```

{
  "total_revision_events": 2,
  "narrowed_or_withdrawn": 1,
  "reaffirmed_unchanged": 1,
  "extended_new_material": 0,
  "constraint_ratio": 0.5,
  "reaffirmed_unchanged_cases": [
    {
      "paper": "the_audit_that_missed_itself",
      "trigger": "Belsky et al. (2022) misattribution recurring, uncorrected,
        in a later paper presented as this corpus's most rigorous

```

```

        self-audit to date"
    }
}
}
Structural flags: none.

```

Two results follow from comparing this independent coding to Section IV's original. First, on the one case that matters most for this paper's credibility, the two codings agree exactly: the Belsky et al. (2022) reaffirmed-unchanged event, coded independently and without reference to the original dataset, describes the identical trigger the original dataset already reported. This is the strongest form of confirmation this paper's method can produce for that specific finding, precisely because it was not assembled by the same process being checked.

Second, the independent coding found one revision event the original dataset does not contain. Falsification condition F-Guillotine2 in *The Audit That Missed Itself* cites RiddleBench's finding that a specific model — Qwen QwQ 32B, scoring 50.86 percent overall, sixth of ten models the benchmark tested — failed to catch its own reasoning errors in 67.7 percent of self-review trials. The paper's own working abstract, during drafting, had generalized this finding to "leading models" before the specific model tested was checked against RiddleBench's own reported rankings and the claim was narrowed to name the model actually measured. This is a genuine narrowing event, dated to the same drafting process that produced the paper's currently published text, and it was not present in Section IV's dataset for this object — not because Section IV's coding was dishonest, but because the event postdates whatever coding pass produced Section IV's eleven-event count, illustrating exactly the kind of update Section III's new *diff_report* function exists to make visible rather than silently absorb.

Extending Section IV's programme-level count by this one newly surfaced, independently sourced event — holding the other six objects' six-events-and-one-reaffirmed-case fixed, since this revision did not have primary access to recode them — moves the totals from eleven events (seven narrowed or withdrawn, one reaffirmed, three extended) to twelve (eight narrowed or withdrawn, one reaffirmed, three extended), and the constraint ratio from 0.636 to 0.667. This paper states plainly what this update does and does not show. It shows that at least one object in the dataset, checked by a process independent of the one that produced Section IV's numbers, reproduces the existing finding exactly and adds a real, dated, additional data point in the same direction the existing finding already pointed. It does not show that the other six objects would survive the same independent check, because this revision did not have primary knowledge sufficient to perform it on them, and Section VI restates this limit rather than lets a reader extrapolate from one verified case to seven.

VI. WHAT THE OUTPUT SHOWS, AND THE ONE THING IT CANNOT VERIFY

The 0.636 constraint ratio — 0.667, extended by Section V's independently confirmed update — is genuine evidence, at the scale of seven papers and twelve events, that this corpus's recent production narrows and withdraws claims more often than it merely reasserts them — a real, if small-sample, data point in favor of Lakatos's progressive rather than degenerating pole, and a real, if small-sample, data point recording exactly one case on the concerning side, reported by the tool rather than omitted. Neither finding demonstrates that Mortality Asymmetry Theory, or any other substantive claim this corpus makes, is correct. The sandbox checks structure — does an exit condition exist, does it point outward, does the revision record show real movement — not truth.

The more specific limitation, and the reason this paper cites Strathern (1997) rather than leaving the point implicit, is this: every falsification condition's *references_external_source* flag, and every revision event's narrowed-versus-reaffirmed classification, was set by a coder — for six of the seven objects in this dataset, by the same collaborative process that wrote the papers being coded, and Section V's independent check does not extend to them. Strathern's paraphrase of Goodhart's observation, developed for a different audit context but exactly on point here, states the risk precisely: when a measure becomes a target, it ceases to be a good measure. Nothing in *sandbox.py*'s logic can detect whether a *references_external_source: true* flag was assigned because the condition genuinely points outward, or because a coder wanted the paper to pass Order 2. Section V demonstrates the tool surviving this test for one object. It is not evidence the other six would, and this paper does not present a partial check as a completed one.

VII. OBJECTIONS

“If the tool cannot verify its own input coding, what does it actually add over just reading the papers directly?”

Compression and consistency, not verification. A reader checking seven papers by hand for whether each has an externally checkable exit condition would reach the same conclusion *sandbox.py* reaches in under a second, but the tool makes the check repeatable, and — because the code is a separate, inspectable artifact rather than prose asserting a conclusion — makes the coding decisions themselves available for a second party to contest line by line, which prose alone does not offer as directly. Section V is this objection answered concretely rather than only in principle: a second party did contest one object's coding, by recoding it independently, and the artifact made that contest possible in a way a paragraph of prose describing the same finding would not have.

“Isn't a single reaffirmed-unchanged case out of eleven — or twelve, after Section V — too small a sample to support any claim about the corpus's general tendency?”

Yes, and Section VI states this directly rather than rounding up. The ratio is reported as what seven papers and twelve events show, not as a corpus-wide finding; F-Sandbox2 states exactly what would be needed to extend the claim responsibly.

“Doesn't building a tool that reports your own corpus's error favorably, by including the embarrassing case, risk simply looking rigorous without being rigorous?”

The specific test is whether the embarrassing case was findable independently of this paper. It was, twice now: *The Audit That Missed Itself* already documented the Belsky recurrence in its own right, before this paper existed, as that paper's own central cautionary finding, and Section V's independent recoding — performed without reference to Section IV's dataset — found it again from a separate starting point. This paper's contribution is not discovering the case. It is showing that a general-purpose structural tool, pointed at real data, surfaces it without being told where to look, and that a second, independent look finds the same thing plus one more.

“Section V only independently recoded one object out of seven. Doesn't presenting an updated 0.667 alongside the original 0.636 risk implying a stronger validation than actually occurred?”

This is a fair concern about presentation, and Section V and Section VI both state the scope of what was checked rather than let the two numbers sit side by side unexplained. The update is reported precisely because it is traceable to a named, dated, independently sourced event — not because six-sevenths of the dataset has now been confirmed, which it has not.

VIII. FALSIFICATION CONDITIONS

F-Sandbox1 (the schema is not actually general): if the research-object schema is shown to be inapplicable, without significant modification, to a paper outside this corpus's own production — for instance, a published paper in an unrelated field — Section II's implicit claim that this is a general-purpose tool rather than a corpus-specific one should be withdrawn.

F-Sandbox2 (the worked example does not generalize): if a substantially larger sample — the remainder of this corpus's catalogue, coded by the same process — produces a constraint ratio materially different from 0.667, or surfaces additional reaffirmed-unchanged cases beyond the one already known, Section VI's characterization of this as a small but genuine positive data point should be revised in whichever direction the fuller sample points.

F-Sandbox3 (the coding is not, in fact, honest): partially tested by Section V, and narrowed accordingly rather than closed. Independent recoding of one of the seven objects, performed by a review process without reference to the original coding, reproduced the original's Order 2 pass rate and reaffirmed-case finding for that object exactly, and surfaced one additional event in the same direction. This condition remains fully live for the other six objects, which no independent process has yet recoded; F-Sandbox3 should be considered closed only once a comparable independent check has been run against the full dataset, not the one-seventh of it Section V was able to verify from primary knowledge.

F-Sandbox4 (the Order 1/Order 2 distinction is not doing real work): if papers that clearly should fail Order 2 by any independent reader's judgment are shown to pass it under this schema, or vice versa, the schema's operationalization of *The Third-Order Audit's* own categories should be revised.

IX. CONCLUSION

A sandbox is not a metaphor once code exists that runs against real data and returns a result its author did not choose in advance, and it is not fully audited once one edition of the paper says so — it is audited to the extent a second, independent pass through the same code, on the same claim, returns the same answer. This revision ran that second pass on the one object it could check from primary knowledge rather than from the corpus's own report of itself, and the two passes agreed, then the second pass found one thing the first had not yet recorded. Six papers still pass on the strength of a coding process this revision did not independently repeat; one is still flagged, correctly, for a real and previously unstated gap; twelve revision events now split 8–3–1 across narrowing, extending, and the one reaffirmed case this corpus already knew was its weakest documented moment, plus one narrowing event this corpus did not know it had recorded until a second process went looking. None of this is offered as evidence that the corpus's substantive claims are true, and Section VI states directly the one thing the tool structurally cannot check at six-sevenths of its own current scale: whether the hands that coded that remaining input were honest ones. A measure that became a target would stop being a good measure the moment it did. This revision's closing position is unchanged from the first: the tool is worth having anyway, precisely because when someone other than its builder ran it, it held.

References

- Strathern, M. (1997). 'Improving ratings': audit in the British University system. *European Review*, 5(3), 305–321.
- Huynh, G. B. (2026). *The Third-Order Audit: A Synthesis Module for Organized Skepticism Across Eight Traditions, and Why Statelessness, Not a New Epistemology, Is Its Only Novel Attribute*.

Huynh, G. B. (2026). *The Third Break: What Mythos 5's Rationalized Non-Compliance Adds to Where Structural Fit Ends*.

Huynh, G. B. (2026). *The Audit That Missed Itself: P0-P1-P2 as a Disclosure Protocol for AI-Assisted Research*.

Huynh, G. B. (2026). *The Ceiling of One Reader; The Right Side of the Guillotine; The Load-Bearing Test; The Prototype and the Update*.

Accompanying Artifacts

sandbox.py — reference implementation, independently runnable, extended in this revision with the `ResearchObject/FalsificationCondition/Revision/ResearchSandbox` classes and JSON loader implied but not published in the first edition, plus a new *diff_report* function.

corpus_data.json — the seven-paper dataset used in Section IV, coded from each paper's own stated text.

independent_recoding.json — the one-object dataset produced for Section V's independent check, coded without reference to *corpus_data.json*.